The unrestricted optimum after Conjecture 20: the one-dimensional mechanism, a width theorem, and what can and cannot be proved

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Abstract

We tried to prove that the unrestricted maximum margin equals the paper's prediction, either the interference-free (IF) value $\nu^{1-s}B_p\Gamma_s(p)$ or the price–programme value $\pi_{n,b}(p)B_p\Gamma_s(p)$ of the companion note. The outcome is as follows. (1) A universal proof of the IF value is impossible. At $p = 3$, $b = 2$ we identify the mechanism behind the counterexamples exactly: a *shared-sign, one-dimensional* family whose feature field is a rotating phase times a real exponential sum. Searching this family (exact LP for the weights, outer search over m real slopes) gives new certified networks with $\gamma/\gamma_{\mathrm{IF}} = 1.063, 1.193, 1.184, 1.457, 1.617, 1.595, 2.011, 2.212$ for $n = 11, \dots, 18$. The excess grows with n. (2) We prove a *width theorem*: a slot-symmetric single-feature network whose input vectors share a sign pattern needs at least $\nu - \lfloor \nu/p \rfloor$ neurons. Every network in (1) attains this bound. The saving that drives the counterexamples is therefore at most a $1/p$ fraction of the neurons, and is zero when $p > \nu$, which covers the paper's experimental range. (3) For $p = 5, 7$ the same mechanism, run at its minimal width, loses to IF by a factor that grows with n (efficiency ≤ 0.19 for $p = 5$, $n \le 16$, and decreasing). Releasing it into the full slot-symmetric class does not help. (4) Decoupling, the remaining step of the price–programme conjecture, is *not* automatic: at $p = 5$ a mean-zero frequency-2 field carried by the same neurons raises a feature's margin by factors 1.2–2.6. In every case we found it still stays $\ge 4\times$ below IF. So the correct universal statement is the price–programme form, and the open content is concentrated in two precise questions for $p \ge 5$.

1 Setting and notation

As in the paper and the companion note: p odd prime, $\rho = e^{2\pi i/p}$, $\theta := \pi/p$, $\nu = n + 1$, $s = \nu/b$. A *single-feature* network has output vectors at frequency 1 only; its margin is $h = \kappa_{\min}(p) \min_x \mathrm{cm}(\Phi(x))$, where $\Phi(x) = \sum_i W_i \prod_t u_{it}(x_t)\rho^{x_t}$ and $\mathrm{cm}(z) = \mathrm{Re}\, z - \cot\theta\, |\mathrm{Im}\, z|$ (companion note, Lemmas 1–2). So $h > 0$ iff every value $\Phi(x)$ lies in the open sector $|\arg z| < \theta$. A network is *slot-symmetric* if $u_{it} = u_i$ for all t; then Φ depends on x only through the count vector $N \in \mathbb{Z}^p_{\ge 0}$, $N_a = \#\{t : x_t = a\}$:

$$\Phi(N) = \sum_{i=1}^{m} W_i \prod_{a \in \mathbb{Z}_p} \big(u_i(a)\rho^a\big)^{N_a}.$$

Efficiency means $\gamma/\gamma_{\mathrm{IF}}$ at $b = 2$ (for single-feature networks at $p \ge 5$, $n \ge 3$ this is the same as the ratio of the single-feature margin to the IF single-feature margin, because $K^*(s, p) = 1$).

2 Shared-sign networks and their fields

Definition 1. A slot-symmetric network has a *shared sign pattern* $\varepsilon \in \{\pm 1\}^p$ if $\varepsilon_a u_i(a) \geq 0$ for all i, a. The *sine pattern* is $\varepsilon_a = +1$ for $0 \leq a < p/2$ and -1 for $p/2 < a < p$, with exponents $e_a := 2a$ resp. $2a - p$, a permutation of $\{0, \ldots, p-1\}$ satisfying $\varepsilon_a \rho^a = e^{i\theta e_a}$. The *one-dimensional (1-D) family* consists of the sine-pattern networks with $u_i(a) = \varepsilon_a e^{\lambda_i e_a}$ (up to scale), $\lambda_i \in \mathbb{R}$.

Lemma 2 (fields)**.** *For a sine-pattern network, $\Phi(N) = e^{i\theta\tau(N)} S(N)$ with $\tau(N) = \sum_a e_a N_a \in \{0, \ldots, (p-1)n\}$ and $S(N) = \sum_i W_i \prod_a |u_i(a)|^{N_a}$, an exponential sum with positive bases. In the 1-D family*

$$\Phi(N) = e^{i\theta\tau} \sum_{i=1}^{m} V_i e^{\lambda_i \tau}, \qquad \tau = \tau(N),$$

so the margin condition is a condition on the $L+1$ values $\tau = 0, \ldots, L$, $L = (p-1)n$, of a rotating phase times a real-exponential sum.

Proof. $u_i(a)\rho^a = |u_i(a)|\varepsilon_a \rho^a = |u_i(a)| e^{i\theta e_a}$; take the product over a with exponents N_a. □

At $p = 3$ the certified counterexample of the earlier notes ($n = 11$, width 8) is, to three digits, a 1-D network: all eight input vectors have the sign pattern $(-, +, +)$ (a translate of the sine pattern), and $\log|u_i(a)| - \log|u_i(0)| = \lambda_i \cdot (0, -1, 1)$ with ratio -1.00 ± 0.03. Re-optimising it inside the 1-D family raises its efficiency from 1.0544 to 1.0631.

3 A width theorem

Theorem 3 (width)**.** *Let p be an odd prime, $n \geq 1$, and let a slot-symmetric single-feature network with a shared sign pattern have positive margin. Then its width is*

$$m \;\geq\; \nu - \left\lfloor \frac{\nu}{p} \right\rfloor.$$

Proof. Positive margin is an open condition, so we may perturb zero entries of the u_i to small values of the prescribed sign. Hence assume all $u_i(a) \neq 0$. Put $d = (p-1)/2$. The map $a \mapsto a + d$ is a cyclic permutation of $\mathbb{Z}_p$ of odd length, so ε cannot alternate along it, and there is a symbol b with $\varepsilon_b = \varepsilon_{b+d}$. Let $a = b + d$ and consider the $n+1$ count vectors $N(k) = (n-k)\mathbf{e}_b + k\mathbf{e}_a$, $k = 0, \ldots, n$. With $r_i = u_i(a)/u_i(b) > 0$,

$$\Phi(N(k)) = C\left(e^{i(\pi-\theta)}\right)^k S(k), \qquad S(k) = \sum_i c_i r_i^k,$$

for constants $C \in \mathbb{C} \setminus \{0\}$, $c_i \in \mathbb{C}$, because $\rho^d = e^{i(\pi-\theta)}$. Positive margin gives $\arg S(k) \in I_k := (\omega_k - \theta, \omega_k + \theta)$ with $\omega_k = -\arg C - (\pi - \theta)k$. Note $\omega_k \equiv -\arg C + \theta k \pmod{\pi}$.

For $\psi \in \mathbb{R}$ let $h_\psi(k) = \mathrm{Im}(e^{-i\psi} S(k)) = \sum_i \mathrm{Im}(e^{-i\psi} c_i)\, r_i^k$. After merging equal bases this is a real exponential sum with at most m terms. By Laguerre's rule of signs it has at most $m - 1$ real zeros unless it vanishes identically, so the sequence $h_\psi(0), \ldots, h_\psi(n)$ has at most $m - 1$ sign changes. Fix a residue k_0 and take $\psi = -\arg C + \theta k_0$. The arc I_k meets the line $\psi + \pi\mathbb{Z}$ iff $\theta(k - k_0) \in (-\theta, \theta) + \pi\mathbb{Z}$, i.e. iff $k \equiv k_0 \pmod{p}$. Call these k *undetermined*. For every other k the sign of $h_\psi(k)$ is fixed by the side of the line on which I_k lies.

- If $k, k+1$ are both determined, $I_{k+1} = I_k - \pi + \theta$; an arc of length 2θ on one side, rotated by $-\pi + \theta$ and still avoiding the line, lies on the other side, so the signs differ.

- If k is undetermined and $0 < k < n$, then $k \pm 1$ are determined ($p \geq 3$), and with I_k centred on the line, I_{k-1} and I_{k+1} are the arcs of length 2θ immediately on either side of it, so $h_\psi(k-1)$ and $h_\psi(k+1)$ again have opposite signs.

So the determined values alternate in sign and there are $|D| - 1$ sign changes, where D is the set of determined k. $h_\psi \not\equiv 0$, since otherwise every $S(k)$ would lie on the line, contradicting $D \neq \emptyset$. Choosing k_0 in a least populated residue class of $\{0, \ldots, n\}$ gives $|D| = n + 1 - \lfloor \nu/p \rfloor$, hence $m - 1 \geq n - \lfloor \nu/p \rfloor$. □

Corollary 4. *(i) For $p > \nu$ a shared-sign single-feature network needs at least ν neurons, as many as the IF block. (ii) For every p, the width saving is at most $\lfloor \nu/p \rfloor$. For networks with equal amplitudes at $b = 2$, the width factor $(\nu/m)^{s-1}$ in the cost comparison with the IF block is therefore at most $\big(\nu/(\nu - \lfloor \nu/p \rfloor)\big)^{s-1} \leq \big(p/(p-1)\big)^{s-1}$: up to $(3/2)^{s-1}$ at $p = 3$, but at most $(5/4)^{s-1}$ at $p = 5$. (iii) Every certified counterexample in Table 1 attains the bound with equality.*

Remark 5. The paper's experiments use $\nu \leq 5$ and $p \geq 11$, so by (i) the mechanism behind the counterexamples has no width advantage anywhere in that range. Width is not the only lever: per unit cost, a flat sign vector has $2^{n/2}$ times the raw output of a cosine input. But numerically this lever alone never sufficed (Section 5).

4 The $p = 3$ optimum: reduction and certified values

Reduction. In the 1-D family, for fixed slopes $\lambda_1, \ldots, \lambda_m$ the cone constraints at $\tau = 0, \ldots, L$ are linear in $(\operatorname{Re} V_i, \operatorname{Im} V_i)$, and the cost amplitude of neuron i is $\sqrt{p/2}\,|V_i|\,Z(\lambda_i)^n$ with $Z(\lambda)^2 = \sum_{k<p} e^{2\lambda k}$. Maximising the margin under a weighted ℓ_1 budget is therefore a second-order cone programme (an LP after polygonal approximation of $|V_i|$). The concave $\ell_{2/\nu}$ cost is handled by reweighting; in practice the optimal amplitudes are nearly equal and uniform weights suffice. The whole problem reduces to choosing m real slopes. We search them with CMA-ES and continuation in n. All values below are then re-verified independently.

Structure of the optimum. Each term sits at the cost-optimal tilt for the stretch of τ it covers. The cost amplitudes agree to within $\sim 10\%$, consecutive terms have nearly antipodal phases ($\approx 172^\circ$ apart at $n = 11$), and the margin is active at most τ. The field is a 10^{-3}–10^{-4} residue of terms up to 10^3 times larger, kept inside the 60° sector at every count class.

n	11	12	13	14	15	16	17	18
width m (bound $\nu - \lfloor \nu/3 \rfloor$)	8 (8)	9 (9)	10 (10)	10 (10)	11 (11)	12 (12)	12 (12)	13 (13)
$\gamma/\gamma_{\mathrm{IF}}$	1.06308	1.19335	1.18360	1.45687	1.61704	1.59546	2.01105	2.21197
previously certified	1.0544	–	1.1091	1.4293	–	–	–	–

Table 1: $p = 3$, $b = 2$. All values are certified by `certify_1d.py`: weights are rounded to ten significant digits, the margin is computed in exact rational arithmetic from the logits over all count vectors (hence over all 3^n inputs), and $\gamma/\gamma_{\mathrm{IF}}$ is enclosed by interval arithmetic.

Hence $\pi_{n,2}(3)\,\nu^{s-1}$ is at least 1.06, 1.19, 1.18, 1.46, 1.62, 1.60, 2.01, 2.21 for $n = 11, \ldots, 18$. The interference-free prediction underestimates the $p = 3$ optimum by a factor that, numerically, grows by about 10% per unit of n.

5 $p \geq 5$: the mechanism loses

Gain/loss accounting. For a 1-D network with nearly equal amplitudes, $\gamma/\gamma_{\mathrm{IF}} = \zeta \cdot 2^{n/2} \cdot (\nu/m)^{s-1}$. Here $2^{n/2}$ is the raw advantage of flat sign vectors over cosines, $(\nu/m)^{s-1}$ the width saving (capped by Theorem 3), and ζ the fraction of the mid-range envelope that survives cancellation. The available gain per unit of n is $\frac{1}{2}\log\frac{2p}{p-1}$: 0.55 at $p=3$, 0.46 at $p=5$, 0.42 at $p=7$. Numerically, $\log\zeta$ falls by 0.52–0.6 per unit of n, so the balance is positive only at $p=3$.

n	6	7	8	9	10	11	12	13	14	15	16
$p=5$	.192	.179	.166	.147	.132	.127	.126	.120	.108	.097	.093
width (bound)	6 (6)	7 (7)	8 (8)	9 (8)	10 (9)	10 (10)	11 (11)	12 (12)	13 (12)	14 (13)	14 (14)
$p=7$	.126	.107	.089	.076	.066	.063	.058	.049	.044	.035	–

Table 2: Best 1-D efficiency $\gamma/\gamma_{\mathrm{IF}}$ found (slope search with exact LP, continuation in n with insertion and deletion moves and fresh global candidates at every n).

The 1-D restriction is not the bottleneck. At $p=5$, $n=10$, released into the full slot-symmetric single-feature class (arbitrary input vectors, `cone.py`), the 1-D optimum 0.1318 is a local optimum (trial with no noise: 0.1318; with noise: lower). Mixing in positive-input neurons, which keeps the field a function of τ but breaks the shared sign pattern, lowered the efficiency at $p=3$, $n=11$ ($1.063 \to 0.82$).

6 Cross-feature compensation: decoupling is not automatic

At $p \geq 5$ several frequencies coexist, and the margin condition at x is that the vector $(\Phi_l(x))_l$ lies in a polyhedron, not that each $\Phi_l(x)$ lies in its own sector. Concretely, at $p=5$ a frequency-2 value $\Phi_2 \approx 0.25|\Phi_1|e^{-i18^\circ}$ makes Φ_1 admissible at $\arg\Phi_1 = 60^\circ$, the $p=3$ sector. The same neurons can produce such a Φ_2 almost for free: the field is a 10^{-3} residue of much larger terms, so a frequency-2 output component with $\sim 10^{-3}$ relative weight, costing $\sim 10^{-6}$ of the budget, gives a Φ_2 of the right size. It is limited only by geometry: its phase follows a different linear functional of the counts.

We tested this with an exact LP over output vectors spanning frequencies 1 and 2, at the slopes of the best 1-D networks, on all count classes and all four shifts δ. Measured gains over frequency-1 outputs at the same neurons: $\times 1.17$ ($n=8$), $\times 1.36$ ($n=12$), up to $\times 2.6$ at non-optimal widths. With a CMA search over slopes for the joint problem: 0.197 ($n=8$) and 0.173 ($n=12$). The frequency-2 output carries 0.2–3% of the output energy. So the decoupling half of the price–programme conjecture (companion note, Conjecture 20′) cannot be proved feature by feature: one feature can live outside its own sector where another, interference-only, field compensates. In every configuration we tried, the compensated feature still stays $\geq 4\times$ below IF.

7 What is proved, what is false, what is open

Statement	Status
$\gamma_b^* = \nu^{1-s} B_p \Gamma_s(p)$ for all p, n, b (Conjecture 20)	**false**: $p = 3$, $b = 2$, $n \geq 11$; excess $2.21\times$ certified at $n = 18$, growing
$\nu^{1-s} B_p \Gamma_s \leq \pi_{n,b}(p) B_p \Gamma_s \leq \gamma_b^* \leq B_p \Gamma_s$	proved (companion note)
$\gamma_b^* = \pi_{n,b}(3)\, B_3 \Gamma_s(3)$ at $p = 3$	proved (one frequency)
width $\geq \nu - \lfloor \nu/p \rfloor$ for shared-sign networks	proved (Theorem 3)
single-frequency networks cannot beat IF when $k_p \geq k^*(n, s)$	proved (companion note); k^* numerical
$\gamma_b^* = \pi_{n,b}(p) B_p \Gamma_s(p)$ (Conjecture 20′, decoupling)	open; genuine obstacle exhibited (Section 6)
$\pi_{n,b}(p) = \nu^{1-s}$ for $p \geq 5$	open; every mechanism tested loses, by a factor growing in n

Answer to "prove that the unrestricted optimum matches the prediction". In the interference-free form this cannot be done: the statement is false at $p = 3$, and increasingly so as n grows. In the price–programme form it holds at $p = 3$ by definition, and there the object to compute is $\pi_{n,b}(3)$. We have shown that, in the class that contains every counterexample, this reduces to choosing $\nu - \lfloor \nu/3 \rfloor$ real slopes, with the weights given by an LP. For $p \geq 5$ the full statement would follow from two claims:

(Q1) $\pi_{n,b}(p) = \nu^{1-s}$, i.e. a single feature cannot be bought below the IF price. Theorem 3 limits the width saving of the $p = 3$ mechanism to $\lfloor \nu/p \rfloor$ neurons, and to none for $p > \nu$. The circularity theorem of the companion note handles single-frequency inputs when $k_p \geq k^*$. A proof needs a cost-weighted (robust) form of the width theorem for arbitrary sign patterns.

(Q2) Decoupling. It has to control compensation by interference-only fields, which demonstrably raises a feature's margin.

Both hold in all our experiments at $p = 5, 7$, by margins of $4\times$ or more that grow with n.

Files. `certify_1d.py` (rigorous certificate for the 1-D family, any p), `oned_certified/` (parameters of the networks in Table 1), `mechanism_scripts/` (the field model, the LP, the slope search and continuation, the joint two-frequency LP, the release test).